PROBLEM 10

Conservative Force and Potential I

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Work & Potential Energy

A particle of mass m moves under a force $\mathbf{F} = -cx^3\mathbf{i}$, where c is a positive constant. Find the potential energy function. If the particle starts from rest at $x = -a$, what is its velocity when it reaches $x = 0$? In the subsequent motion, where does it stop again?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m moves along the Ox axis subject to a position-dependent non-linear restoring force directed towards the origin.
- **Kinematic Degree of Freedom:** 1 degree of freedom, parametrized by the Cartesian coordinate $x(t) \in \mathbb{R}$.
- **Hypotheses:**

1. The motion is strictly one-dimensional along the x-axis.

2. The particle is treated as an ideal point particle of constant mass m.

3. No dissipative or frictional forces are present (conservative field).

- **Initial Conditions:** At time $t = 0$, the particle is released from rest at position $x(0) = -a$:

$$x(0) = -a, \qquad v(0) = \dot{x}(0) = 0$$

- **Physical Parameters:**

- m: Mass of the particle $[\mathrm{kg}]$.
- c: Positive force proportionality constant $[\mathrm{N \cdot m^{-3}} = \mathrm{kg \cdot s^{-2} \cdot m^{-2}}]$.
- a: Initial displacement amplitude $[\mathrm{m}]$ ($a > 0$).

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Reference Frame:** Let $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be a fixed Cartesian inertial frame with origin at O.
- **Kinematics Vectors:**
- Position: $\mathbf{r}(t) = x(t)\mathbf{i}$.
- Velocity: $\mathbf{v}(t) = \dot{x}(t)\mathbf{i}$.
- Acceleration: $\mathbf{a}(t) = \ddot{x}(t)\mathbf{i}$.
- **Force Vector:**

$$\mathbf{F}(x) = -cx^3\mathbf{i}$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Calculation of the Potential Energy Function $V(x)$

Pedagogical Justification: In a one-dimensional problem where the force depends exclusively on position x, the force field is automatically irrotational (curl is trivially zero in 1D) and conservative. By definition of a conservative force (Notes.pdf Section 4.4, Eqs. 4.14–4.16), the work done by the force corresponds to the negative variation of potential energy, $\mathbf{F} = -\boldsymbol{\nabla} V = -\frac{dV}{dx}\mathbf{i}$. Integrating this relation allows us to replace the second-order nonlinear differential equation $m\ddot{x} + cx^3 = 0$ with an algebraic energy conservation law (first integral of motion).

We define the scalar potential energy $V(x)$ by choosing the reference datum $V(0) = 0$ at the origin:

$$V(x) - V(0) = -\int_0^x F_x(x')\,dx'$$

Substitute $F_x(x') = -cx'^3$:

$$V(x) = -\int_0^x (-cx'^3)\,dx' = c\int_0^x x'^3\,dx'$$

Evaluating the definite integral by the Fundamental Theorem of Calculus:

$$V(x) = c\left[\frac{x'^4}{4}\right]_0^x = \frac{1}{4}cx^4$$

2. Evaluation of the Total Mechanical Energy E

> **Pedagogical Justification:** Since no non-conservative forces act on the particle, the total mechanical energy $E = T + V(x)$ is an invariant of the motion ($dE/dt = 0$). Evaluating E at the initial release state determines the accessible energy surface for all subsequent times.

At $t = 0$, with $x(0) = -a$ and $v(0) = 0$:

$$T(0) = \frac{1}{2}m[v(0)]^2 = 0$$

$$V(-a) = \frac{1}{4}c(-a)^4 = \frac{1}{4}ca^4$$

Therefore, the total mechanical energy is:

$$E = T(0) + V(-a) = \frac{1}{4}ca^4$$

3. Particle Velocity at $x = 0$

> **Pedagogical Justification:** Conservation of mechanical energy equates the initial potential energy to the sum of kinetic and potential energies at any point x. At the origin $x = 0$, the potential energy reaches its absolute minimum $V(0) = 0$, meaning the kinetic energy T is maximized.

Applying conservation of mechanical energy at $x = 0$:

$$E = T(0) + V(0) \implies \frac{1}{4}ca^4 = \frac{1}{2}mv_0^2 + \frac{1}{4}c(0)^4$$

$$\frac{1}{2}mv_0^2 = \frac{1}{4}ca^4$$

Isolate v_0^2:

$$v_0^2 = \frac{2}{m}\left(\frac{1}{4}ca^4\right) = \frac{ca^4}{2m}$$

Taking the square root:

$$|v_0| = \sqrt{\frac{ca^4}{2m}} = a^2\sqrt{\frac{c}{2m}}$$

Direction of motion: Since the particle was released at $x = -a < 0$, the force acting on it for all $x < 0$ is:

$$F_x = -cx^3 = -c(-|x|)^3 = +c|x|^3 > 0$$

Because the force is strictly directed along $+\mathbf{i}$, the particle accelerates towards the right. Consequently, upon reaching $x = 0$, its velocity is directed in the positive direction:

$$\mathbf{v}(0) = a^2\sqrt{\frac{c}{2m}}\,\mathbf{i}$$

4. Subsequent Stopping Point

> **Pedagogical Justification:** A particle stops momentarily when its velocity vanishes, $v = 0$, which corresponds to a classical turning point where the kinetic energy $T = 0$. By energy conservation, this requires $V(x_{stop}) = E$.

Set kinetic energy to zero:

$$T = 0 \implies V(x_{stop}) = E$$

$$\frac{1}{4}cx_{stop}^4 = \frac{1}{4}ca^4$$

Dividing both sides by the non-zero constant $\frac{1}{4}c$:

$$x_{stop}^4 = a^4 \implies x_{stop}^4 - a^4 = 0$$

Factorizing as a difference of squares:

$$(x_{stop}^2 - a^2)(x_{stop}^2 + a^2) = (x_{stop} - a)(x_{stop} + a)(x_{stop}^2 + a^2) = 0$$

Since $x_{stop} \in \mathbb{R}$, $x_{stop}^2 + a^2 > 0$, yielding two real solutions:

$$x_{stop} = -a \quad \text{or} \quad x_{stop} = +a$$

Since the motion started at $x = -a$ and proceeded across $x = 0$ in the $+x$ direction, the *subsequent* stopping point is:

$$x_{stop} = +a$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Symmetry & Anharmonic Oscillation:** The potential $V(x) = \frac{1}{4}cx^4$ is strictly even ($V(-x) = V(x)$) and convex ($V''(x) = 3cx^2 \geq 0$). The particle is trapped in a symmetric potential well, executing periodic, bounded, anharmonic oscillations between $x = -a$ and $x = +a$.
- **Dimensional Analysis:**
- $[c] = \mathrm{N/m^3} = \mathrm{kg \cdot m \cdot s^{-2} \cdot m^{-3}} = \mathrm{kg \cdot m^{-2} \cdot s^{-2}}$.
- $[v_0] = [a^2]\sqrt{[c]/[m]} = \mathrm{m^2}\sqrt{\frac{\mathrm{kg \cdot m^{-2} \cdot s^{-2}}}{\mathrm{kg}}} = \mathrm{m^2}\sqrt{\mathrm{m^{-2} \cdot s^{-2}}} = \mathrm{m^2 \cdot m^{-1} \cdot s^{-1}} = \mathrm{m/s}$.

The velocity dimensions match perfectly.

- **Limit and numerical check:** the energy balance is dimensionally homogeneous, $[c\,a^4] = \mathrm{N \cdot m^{-3} \cdot m^4} = \mathrm{J}$. For $a \to 0$ the release energy vanishes and $v_0 = a^2\sqrt{c/(2m)} \to 0$. An RK4 integration (step 10^{-4}, $m = c = a = 1$ in SI) of $m\ddot{x} = -cx^3$ from $x = -1$ at rest gives $v = 0.707107\ \mathrm{m/s}$ at $x = 0$ against $\sqrt{1/2} = 0.707107\ \mathrm{m/s}$, and the next stop at $x = 0.99999999$ against $x_{stop} = +a = 1$.

Source: Problems.pdf